

MATH1231 CALCULUS

Chapter 1A

Term II 2026

Lecturer:

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Lectures:

Tuesdays 11:00 - 12:00

Wednesdays 13:00 - 14:00

Fridays 15:00 - 16:00



Keith Burrows Theatre

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UNSW Course Outline

MATH1231 Mathematics 1B - 2026

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Assessments

Assessment Structure

Assessment Item	Weight	Relevant Dates
Weekly Lessons Assessment Format: Individual	10%	Start Date: Various. Approximately two weeks before each weekly deadline. Due Date: There is one deadline each Tuesday at 5pm starting from Week 2.
Lab Tests Assessment Format: Individual	30%	Start Date: Not Applicable Due Date: Not Applicable
Assignment Assessment Format: Individual	10%	Start Date: Week 5 Due Date: Week 8 Tuesday 11:59pm
Final Examination Assessment Format: Individual	50%	Start Date: Not Applicable Due Date: Not Applicable

Overview

Ch 1 Functions of Several Variables

Ch 2 Integration Techniques

Ch 3 Ordinary Differential Equations



Taylor Series } **Ch 4**
Sequences and Series }

Averages, Arc Length, Surface Area } **Ch 5**



With an absurd oversimplification, the 'invention' of the calculus is sometimes ascribed to two men, Newton and Leibniz. In reality, the calculus is the product of a long evolution that was neither initiated nor terminated by Newton and Leibniz, but in which both played a decisive part.

— *Richard Courant* —

AZ QUOTES

1. Functions of Several Variables

Most functions depend on more than one variable.

- Example - Euclidean distance from the origin

one-dimensional $d = f(x) = \sqrt{x^2} = |x|$

two-dimensional $d = f(x, y) = \sqrt{x^2 + y^2}$

three-dimensional $d = f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$

- Example - power of wind turbine

$$P(r, v) = C r^2 v^3 ,$$

where $C = \text{constant}$, r is blade length, and v is wind speed.

► Geometrically

$y = f(x)$ represents a curve in $\mathbb{R}^2$

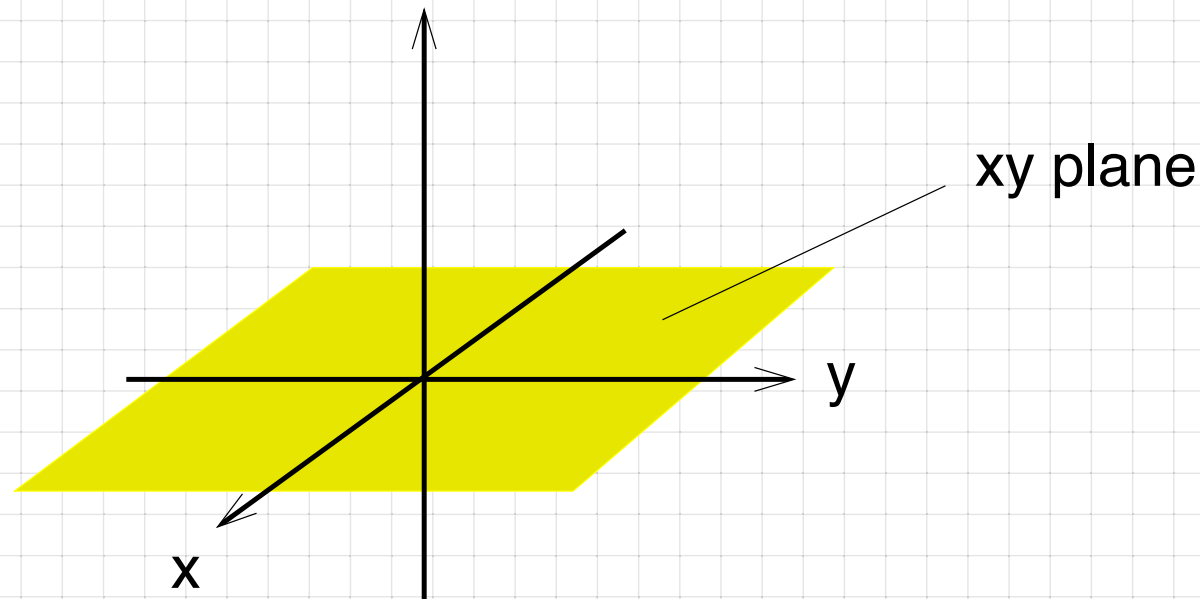
$z = f(x, y)$ represents a surface in $\mathbb{R}^3$

formally this surface is the set $\{(x, y, f(x, y)) : (x, y) \in D\}$,
where D is the domain of f

(1.1) Sketching surfaces in $\mathbb{R}^3$

$$z = F(x, y)$$

Conventional orientation
 z



Level Curves and Profiles

- ▶ A **level curve** of a function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a curve in $\mathbb{R}^2$ defined by $F(x, y) = C$ where C is a constant. In Science and Engineering (and software) these are usually called **contours**.

We can think of them as **horizontal slices** at constant z through the function surface.

- ▶ A **profile** is obtained from a plot of z versus y (with $x = \text{constant}$) or z versus x (with $y = \text{constant}$).

We can think of these as **vertical cuts** through the function surface.

We can combine level curves and profiles to visualize and sketch the surface.

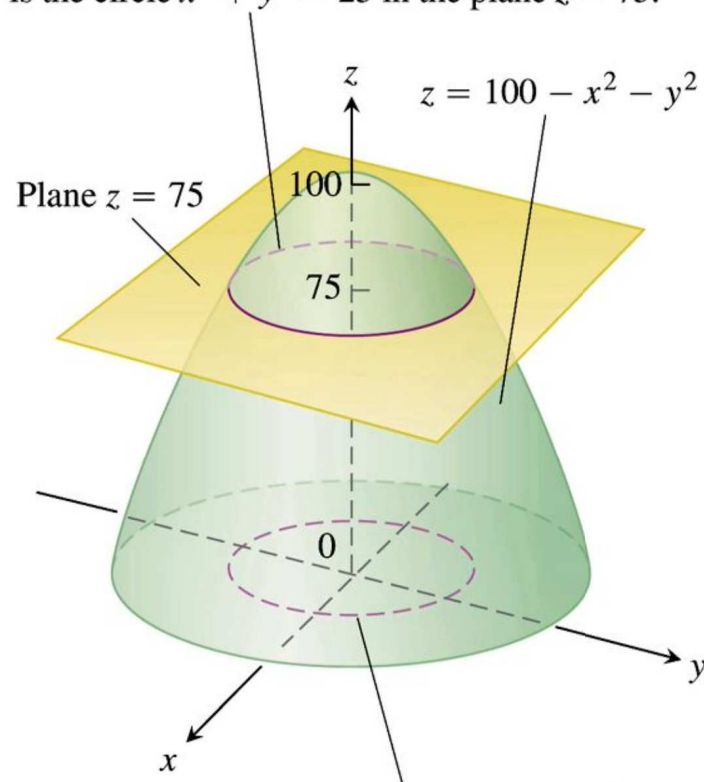
Example

$$z = 100 - x^2 - y^2$$

Example

$$z = 100 - x^2 - y^2$$

The contour curve $f(x, y) = 100 - x^2 - y^2 = 75$ is the circle $x^2 + y^2 = 25$ in the plane $z = 75$.



The level curve $f(x, y) = 100 - x^2 - y^2 = 75$ is the circle $x^2 + y^2 = 25$ in the xy -plane.

Example

$$x^2 + y^2 - z^2 = 1$$

Sketch some level curves and hence sketch the surface in $\mathbb{R}^3$.

Example

$$x^2 + y^2 - z^2 = 1$$

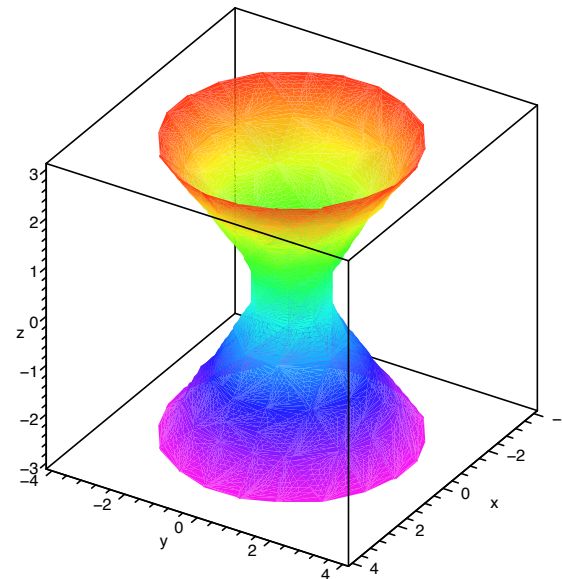
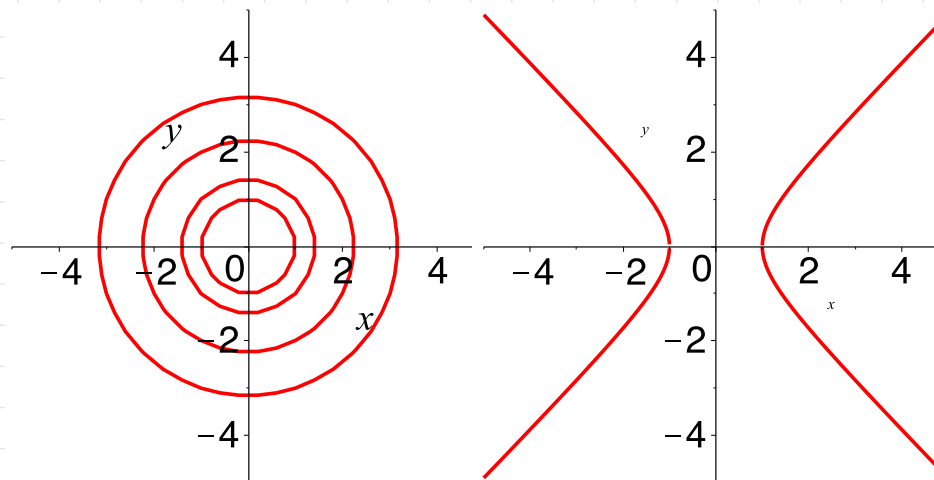
Sketch some level curves and hence sketch the surface in $\mathbb{R}^3$.

$z = F(x, y)$ is not defined explicitly; but use same principles;

level curves with $z = 0$ are circles with radius 1

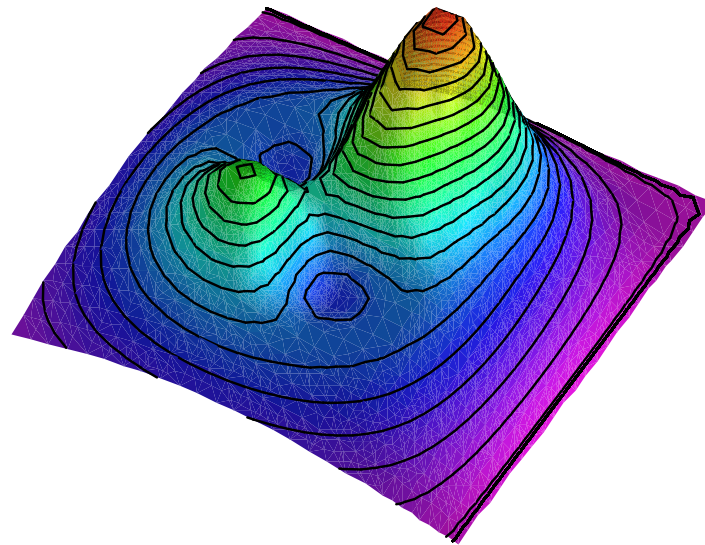
level curves with $z = \pm 1$ are circles with radius $\sqrt{2}$.

level curves with $z = \pm 2$ are circles with radius $\sqrt{5}$.

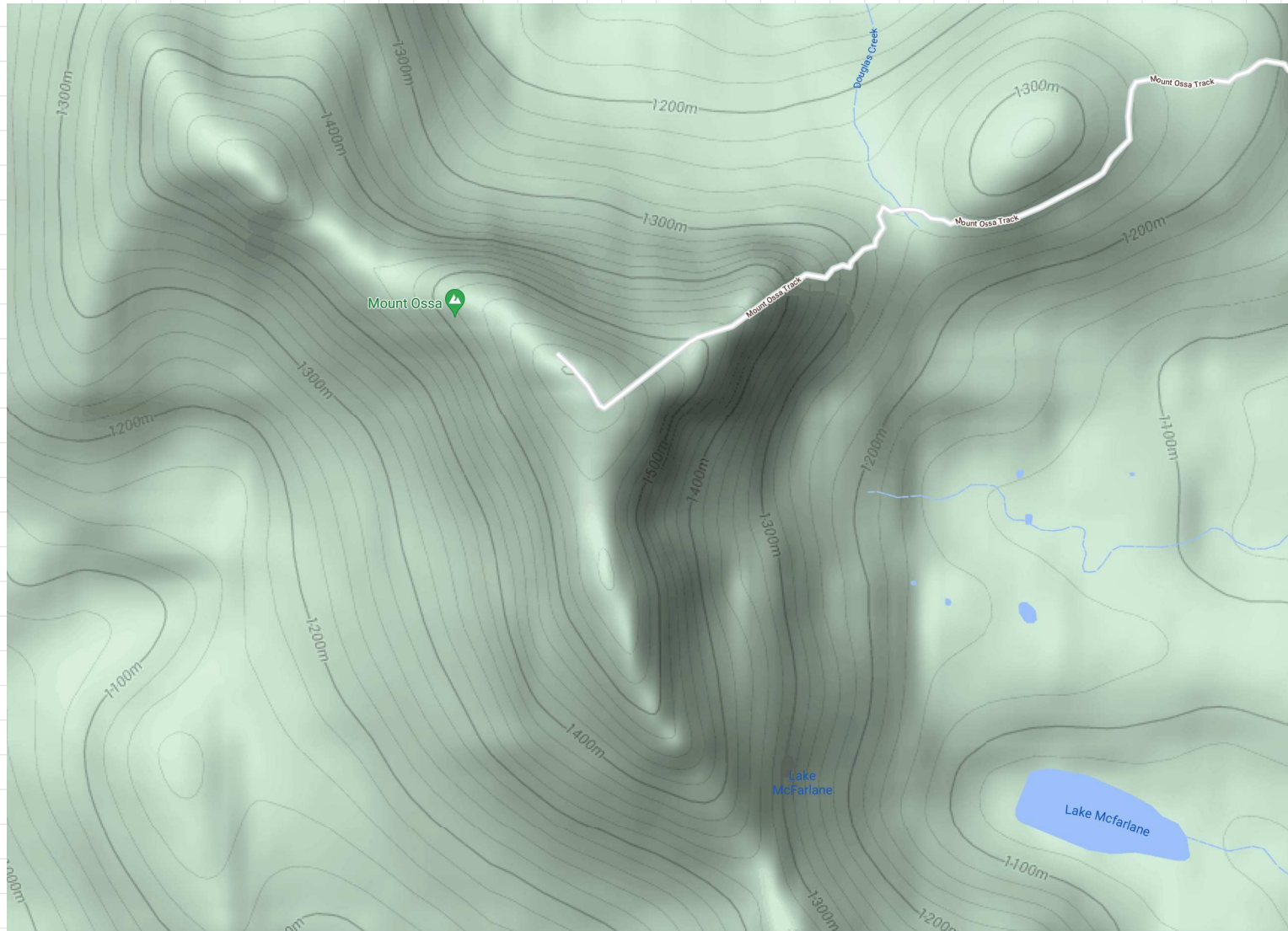


- Tricky example - use MAPLE – plot3d

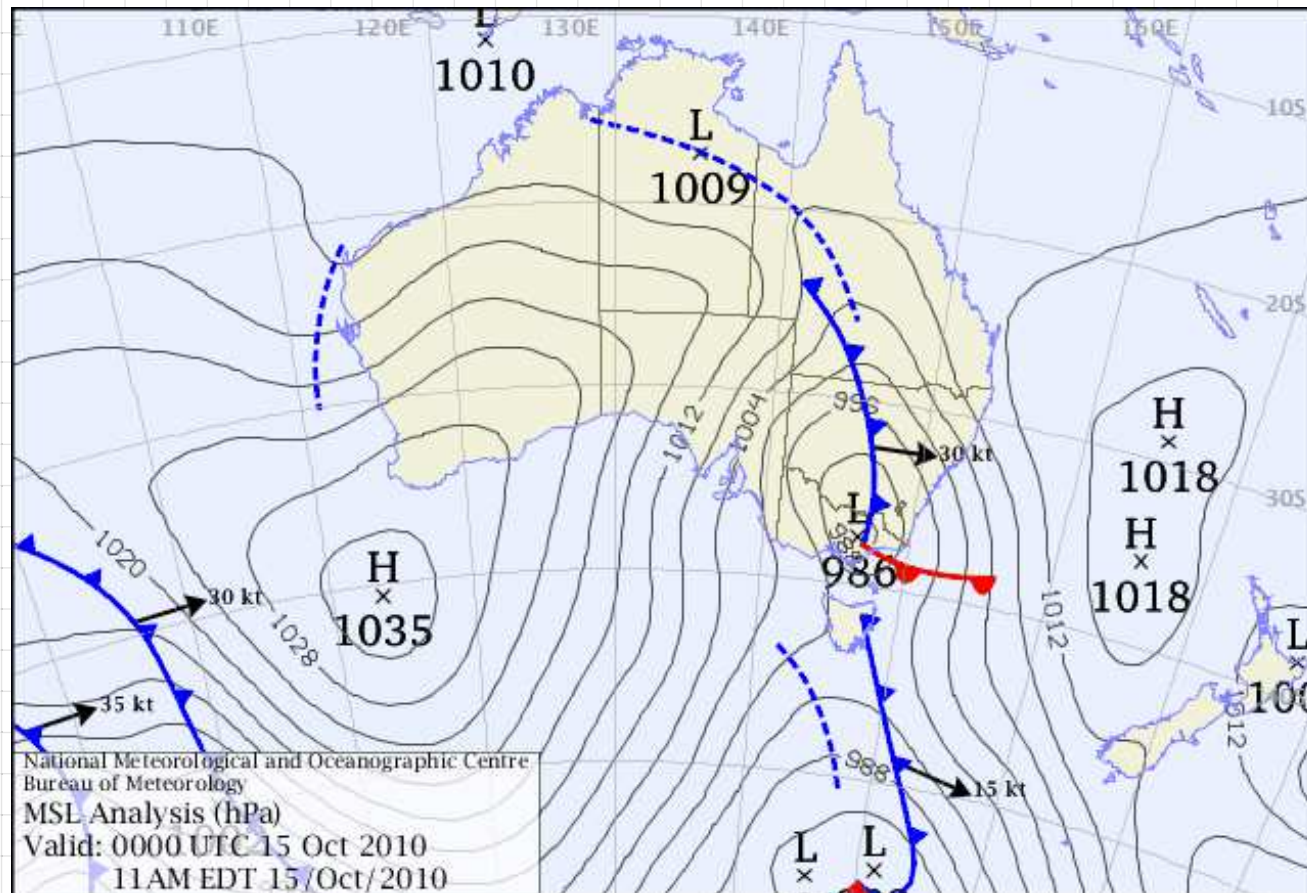
$$z = 2 \cos\left(\frac{2x}{5}\right) \cos\left(\frac{2y}{5}\right) + 5xy \exp(-(x^2 + y^2)) + 3 \exp(-((x-2)^2 + (y-2)^2))$$



Contours on topographic maps:
Level sets of the height of earth's surface above mean sea level!



Contours/Level sets of the mean-sea-level pressure (units of hPa = 100 N/m²)



Partial Differentiation

- ▶ **ordinary derivative** of $f(x)$

- ▶ **partial derivative** of fn of several variables:

Differentiate with respect to one of the variables while holding the other variables fixed.

Given $f(x, y)$, the basic idea is to find the rate of change in f with respect to one variable, while the other variable is held fixed.

Example:

Given $f(x, y) = x^2y + y^4$, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$

Alternate notation:

- ▶ The partial derivative of f w.r.t. x

$$\frac{\partial f}{\partial x} \equiv f_x(x, y) \equiv D_1 f$$

- ▶ The partial derivative of F w.r.t. y

$$\frac{\partial f}{\partial y} \equiv f_y(x, y) \equiv D_2 f$$

Example:

a.) Given $f(x, y) = e^{xy}$, calculate $f_x(1, 1)$ and $f_y(2, 0)$

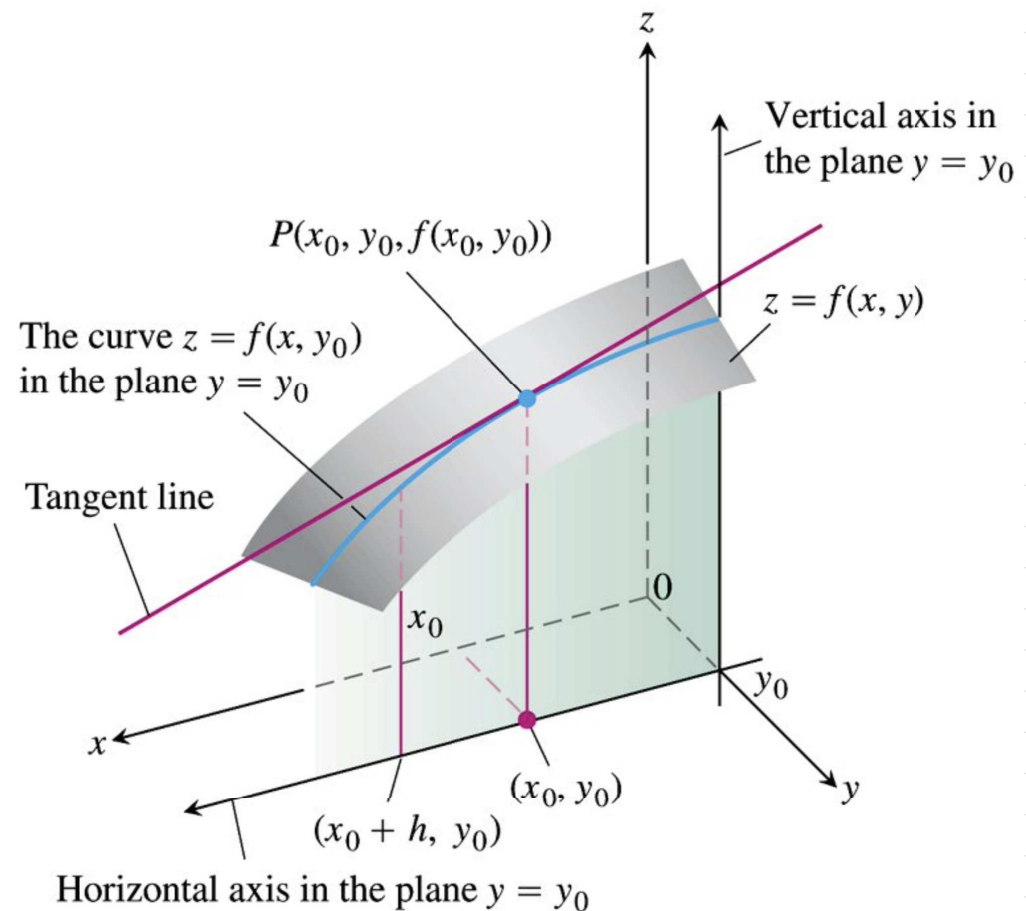
b.) Given $f(x, y) = x^2y \log x + e^{(x^2+y^2)}$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$

Definition of partial derivative

the **partial derivative** of $f(x, y)$ with respect to x at the point (x_0, y_0) is

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

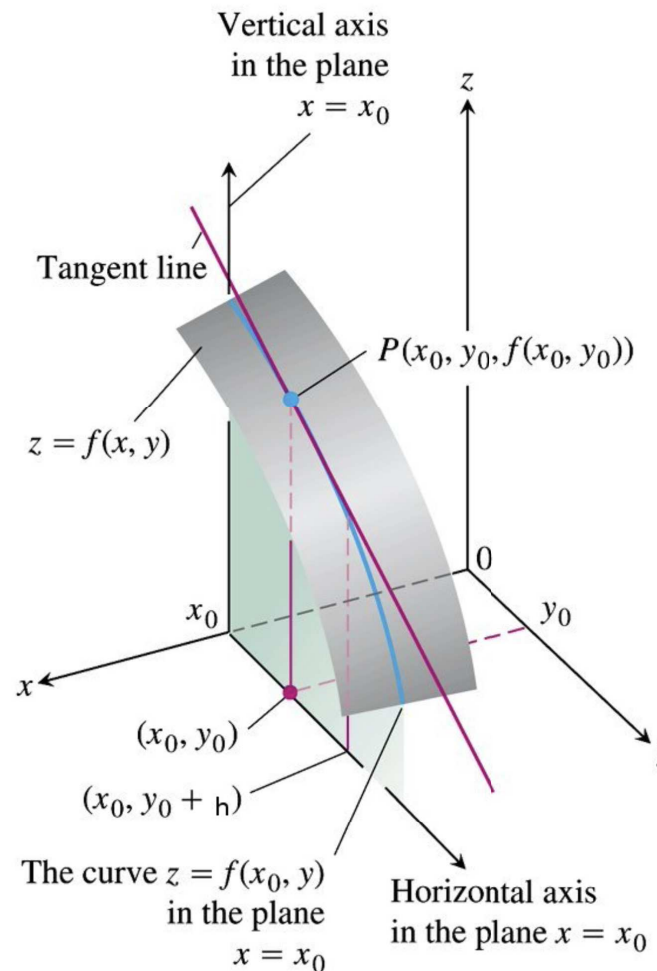
provided the limit exists.



the **partial derivative** of $f(x, y)$ with respect to y at the point (x_0, y_0) is

$$\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h}$$

provided the limit exists.



Geometrical Interpretation

$\frac{\partial F}{\partial x}$ is the slope of the surface $z = F(x, y)$ in the x direction.

$F_x(a, b)$ is the gradient of the tangent to the cross section at (a, b) when the surface $z = F(x, y)$ is intersected with the plane $y = b$.

$\frac{\partial F}{\partial y}$ is the slope of the surface $z = F(x, y)$ in the y direction.

$F_y(a, b)$ is the gradient of the tangent to the cross section at (a, b) when the surface $z = F(x, y)$ is intersected with the plane $x = a$.

Differentiation rules are a natural extension of what you know already:

If $f = f(x, y)$ and $g = g(x, y)$, then

$$\frac{\partial}{\partial x} (f + g) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}$$

$$\frac{\partial}{\partial x} (f \cdot g) = \frac{\partial f}{\partial x} g + f \frac{\partial g}{\partial x}$$

$$\frac{\partial}{\partial x} \left(\frac{f}{g} \right) = \frac{\frac{\partial f}{\partial x} g - f \frac{\partial g}{\partial x}}{g^2} \quad \text{if } g(x, y) \neq 0$$

Practice examples

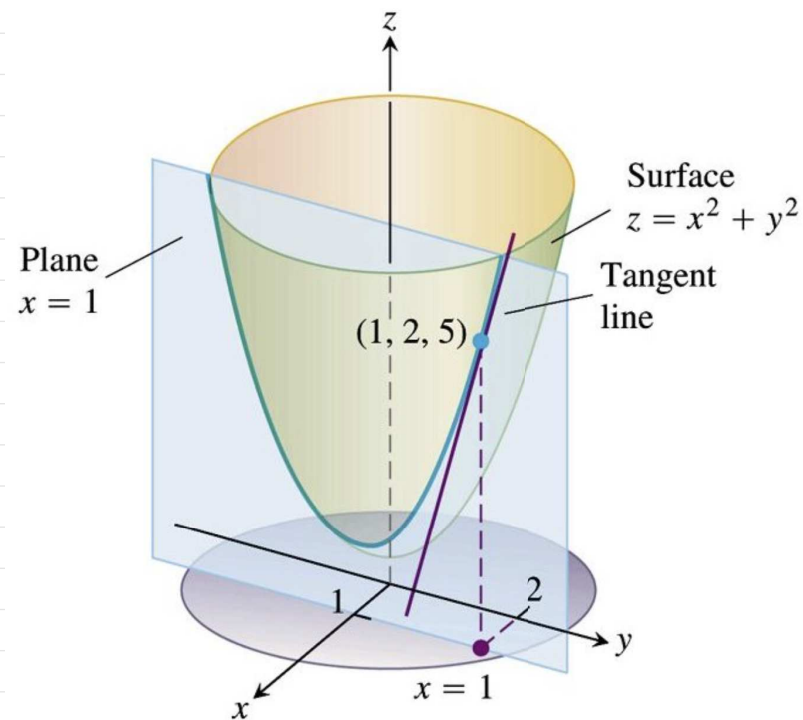
Find $\frac{\partial F}{\partial x}$ and $\frac{\partial F}{\partial y}$

1. $F(x, y) = 3y^2x + \cos(y) + y \sin(x)$

2. $F(x, y) = \frac{e^{xy}}{\sin(y)}$ for $\sin(y) \neq 0$

Example

The plane $x = 1$ intersects the paraboloid $z = x^2 + y^2$ in a parabola as shown in the following diagram. Calculate the slope of that tangent line to the parabola at the point $(1, 2, 5)$.



The tangent to the curve of intersection of the plane $x = 1$ and surface $z = x^2 + y^2$ at the point $(1, 2, 5)$

Higher-order derivatives

Second order partial derivatives are partial derivatives of first order partial derivatives

$$\frac{\partial^2 F}{\partial x^2} \equiv \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial x} \right) \equiv F_{xx} \equiv D_1^2 F$$

$$\frac{\partial^2 F}{\partial y^2} \equiv \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial y} \right) \equiv F_{yy} \equiv D_2^2 F$$

$$\frac{\partial^2 F}{\partial x \partial y} \equiv \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial y} \right) \equiv F_{yx} \equiv D_1 D_2 F$$

$$\frac{\partial^2 F}{\partial y \partial x} \equiv \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial x} \right) \equiv F_{xy} \equiv D_2 D_1 F$$

Warning: The F_{xy} and F_{yx} notation is not standard, but very commonly used in engineering and science

Example

Let $F(x, y) = x^2 \cos(y) + y^2 \sin(x)$. Find $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$, $\frac{\partial^2 F}{\partial x \partial y}$ and $\frac{\partial^2 F}{\partial y \partial x}$.

- ▶ Theorem: If $F = F(x, y)$ is a continuous function of two variables (x, y) and all of its first and second order partial derivatives are continuous then

$$\boxed{\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x}}$$

- ▶ *Proof: See comments in notes. A proper proof (not given) requires a formal treatment of continuity.*
- ▶ *Continuity:*
 $G = G(x, y)$ is continuous at (x_0, y_0) if for every $\epsilon > 0$ there exists a $\delta > 0$ such that $|G(x, y) - G(x_0, y_0)| < \epsilon$ whenever $0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$.

Example

Consider the function $f(x, t) = x^2 + t^2/2$

(a) Calculate f_{xx} and f_{tt}

(b) Use (a) to show that $f(x, y)$ is a solution to the partial differential equation (PDE)

$$f_{xx} - 2f_{tt} = 0 \quad (1)$$

Example $F(x, y) = e^x \sin(xy)$

$$\frac{\partial F}{\partial x} = e^x \sin(xy) + e^x y \cos(xy)$$

$$\frac{\partial^2 F}{\partial x^2} = e^x \sin(xy) + e^x y \cos(xy) + e^x y \cos(xy) - e^x y^2 \sin(xy)$$

$$\frac{\partial^2 F}{\partial y \partial x} = e^x x \cos(xy) + e^x \cos(xy) - e^x yx \sin(xy)$$

$$\frac{\partial F}{\partial y} = e^x x \cos(xy)$$

$$\frac{\partial^2 F}{\partial y^2} = -e^x x^2 \sin(xy)$$

$$\frac{\partial^2 F}{\partial x \partial y} = e^x x \cos(xy) + e^x \cos(xy) - e^x xy \sin(xy)$$